Waves at boundaries

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Describe reflection and transmission at a boundary.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 1.*
Need wave speed; target distance=speed×3/2. The pulse travels to the target and back.

02. Worked example: explain the model
Fixed: inverted reflected pulse; free: same orientation, direction reversed.

03. Read and interpret the model
Downward/inverted pulse travels back along rope.

04. Apply the physics
6 J.

05. Explain your reasoning
No; source fixes frequency. Speed and wavelength may change; amplitude depends energy transfer and medium properties.

06. Review: connect an earlier idea *Review: Unit 5, Day 4.*
Q=mL=2400 J.

07. Exit check
Upright.

08. STRETCH: test the model
Displacements add during overlap; pulses subsequently pass through each other in the linear idealization.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 1.*
Second/first range=5/3≈1.667. Delay does not establish mass.

02. Worked example: explain the model
Fixed: inverted reflected pulse; free: same orientation, direction reversed.

03. Read and interpret the model
Downward/inverted pulse travels back along rope.

04. Apply the physics
Transmitted .50 J; reflected 25%, absorbed 12.5%, transmitted 62.5%.

05. Explain your reasoning
No; source fixes frequency. Speed and wavelength may change; amplitude depends energy transfer and medium properties.

06. Review: connect an earlier idea *Review: Unit 5, Day 4.*
L=|Q|/m≈300000 J/kg; sample Q=-120000 J.

07. Exit check
2 m and 4 m.

08. STRETCH: test the model
Total zero at that point; energy may be in motion/elsewhere and interference redistributes it.